

Note by identical reasoning as on [pages 147-148](#), we can see that a holomorphic function between Riemann surfaces is continuous.

Proposition: If M and N are Riemann surfaces and $F : M \rightarrow N$ is a holomorphic function, then for any complex charts (U, φ) on M and (V, ψ) on N we have that $\psi \circ F \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) on $\varphi(U \cap F^{-1}(V))$

Proof:

Because F is continuous, we know that $U \cap F^{-1}(V)$ is an open subset of M . Now we have nothing to show if $U \cap F^{-1}(V) = \emptyset$. So, we can assume $U \cap F^{-1}(V)$ is nonempty. Then by shrinking U , we can without loss of generality just say that $U \subseteq F^{-1}(V)$. That way, our problem becomes showing that $\psi \circ f \circ \varphi^{-1}$ is a holomorphic function (in the complex analytic sense) from $\varphi(U)$ into $\psi(V) \subseteq \mathbb{C}$.

Fortunately let $x = \varphi(p) \in \varphi(U)$. Then we know there exists complex charts (U', φ') on M and (V', ψ') on N with $p \in U'$, $F(U') \subseteq V'$, and $\psi' \circ F \circ (\varphi')^{-1}$ a holomorphic function (in the complex analytic sense) from $\varphi'(U')$.

But now $\psi \circ f \circ \varphi^{-1} = (\psi \circ (\psi')^{-1}) \circ (\psi' \circ f \circ (\varphi')^{-1}) \circ (\varphi' \circ \varphi^{-1})$ is a holomorphic function from $\varphi(U \cap U')$ to $\psi(V \cap V')$. Since $\varphi(U \cap U')$ is an open set containing x (since $U \cap U'$ is an open set containing p), we've proven that $\psi \circ f \circ \varphi^{-1}$ is holomorphic at $x \in \varphi(U)$. And since x was arbitrary, we can conclude that $\psi \circ f \circ \varphi^{-1}$ is a holomorphic function from $\varphi(U)$. ■